```
#include<iostream>
using namespace std;
class complex{
   private:
   float real;
   float imag;
   public:
   complex(){
      real=0;
      imag=0;

   }
   complex(float r,float i){
      real=r;
      imag=i;
   }
   ~complex(){

   }
   void display(){
      cout<<real<<"+"<<imag<<"i";
   }
   complex operator+(complex c){
      return complex(real+c.real,imag+c.imag);
   }
   complex operator-(complex c){
      return complex(real-c.real,imag-c.imag);
   }
   complex operator*(complex c){
      float r=real*c.real-imag*c.imag;
      float i=real*c.imag+imag*c.real;
      return complex(r,i);

   }
};
int main(){
   complex c1(4,5),c2(2,3);
   cout<<"\nFirst complex Number:";
   c1.display();
   cout<<"second complex Number:";
   c2.display();
   complex sum=c1+c2;
   cout<<"\nsum:";
   sum.display();
```

```
    complex diff=c1-c2;
    cout<<"difference:";
    diff.display();
    complex prod=c1*c2;
    cout<<"product:";
    prod.display();
    return 0;
}

/*
PS D:\swarupa44> cd "d:\swarupa44\" ; if ($?) { g++ complex2.cpp -o complex2 } ; if
($?) {
.\complex2 }
First complex Number:4+5isecond complex Number:2+3i
sum:6+8idifference:2+2iproduct:-7+22i
*/
```